Modify the URL to *http://127.0.0.1:5601/app/sense*. It will display a two-pane window. The left pane is used to format the request for ElasticSearch and the right pane will display the response to the request. In the left pane, type *GET _cluster/health?pretty* and press the small green arrow button as shown in Figure 2.

While you are typing, *Sense* will provide auto completion for APIs. This can also be achieved using *curl* but the *Sense* plugin is a better option.

In the next section, we will use *Sense* to store, search and analyse data.

Storing, searching and analysing data

Storing data: Let us assume that we want data about students to be analysed. We will store the data, using the ElasticSearch RESTful API. We will create a student with the following attributes – first name, gender, city, state, SSC, HSC, graduate degree, goal and hobbies. The attributes SSC, HSC and graduate degree will store percentage as value.

We need to use the PUT request to store [create / update] the data. Use the following command in *Sense* [Figure 3]:

```
PUT /college/students/1
{
  "firstname":"Student1",
  "gender" : "Male",
  "city" : "Mumbai",
  "state" : "Maharastra",
  "ssc": 69,
  "hsc" : 77,
  "grad" : 90,
  "graddegree" :"BCA",
  "goal" : "To work in organization
where my skills are put to use",
  "hobbies" :
["reading","writing","dancing"]
}
```

The act of storing the data in ElasticSearch is called indexing. The diagram shows a student document. For every student, a document will be made. This data is stored in clusters, which can contain many indices.

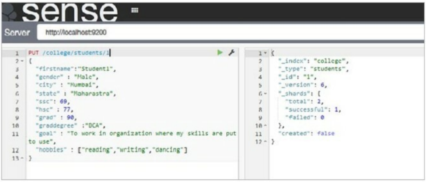

Figure 3: Storing data

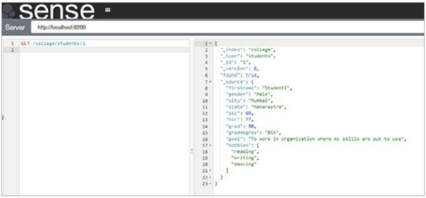

Figure 4: Searching data

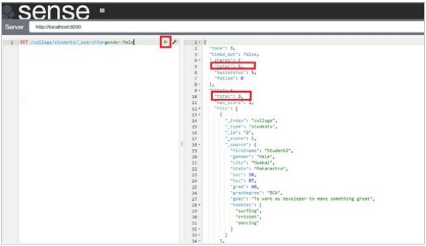

Figure 5: Searching all male students

So each student is a document or rather, we can say that the document is of type 'student'. It is stored in the college index, which in turn is stored in a cluster. When we say */college/student/1* it is interpreted as */index/type/id*. This is clear from Figure 3, which displays *_type*, *_index*, *_id* as a response to the request.

Similarly, let's create some more students so that we can analyse the documents. While adding data, make sure to have variations in gender, state, city, hobbies and goals.

Querying data: To query or search the data stored, the ElasticSearch API uses the GET method followed by the URL. For example, to retrieve the record of Student 1, execute the following command, which is shown in Figure 4.

```
GET /college/students/1
```